WORKED SOLUTION

# STEP III Specimen paper, Question 4

Pure • Tangents to a Parabola

**STEP III Specimen paper, Question 4**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/spec-s3-q4

## The key idea

A line touches a parabola exactly when substituting it into the parabola's equation gives a quadratic with a repeated root. So the first part is a one-line discriminant calculation, and it hands us a family of tangents labelled by $t$. For the second part, think of the tangents from a given point as the values of $t$ satisfying a quadratic: then everything about $L$ and $M$ can be written in terms of the sum and product of the roots, which come straight from the coefficients.

## Solution

### The tangent line

From the line, $x = ty - at^2$. Substituting into $y^2 = 4ax$ gives

$$y^2 = 4aty - 4a^2t^2 \iff y^2 - 4aty + 4a^2t^2 = 0 \iff (y - 2at)^2 = 0.$$

This quadratic in $y$ has the repeated root $y = 2at$, so the line meets the parabola at exactly one point, and does so with multiplicity two: it touches the parabola. (It is not a line parallel to the axis, which would also meet the parabola only once, because it is not horizontal; for $t = 0$ it is the line $x = 0$, the tangent at the vertex.) The corresponding $x$ is $x = t(2at) - at^2 = at^2$, so the point of contact is

$$(at^2,\ 2at).$$

As a cross-check, differentiating $y^2 = 4ax$ implicitly gives $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{2a}{y} = \frac{1}{t}$ at this point, which is the gradient of $ty = x + at^2$.

### The locus

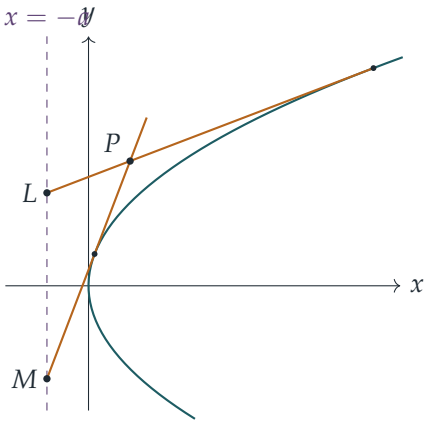
Every point of the parabola is $(at^2, 2at)$ for exactly one $t$, so every tangent is one of the lines $ty = x + at^2$. Let the given point be $P = (h, k)$. The tangent with parameter $t$

passes through $P$ if and only if

$$at^2 - kt + h = 0.$$

For two tangents we need two distinct real roots $t_1, t_2$, that is $k^2 > 4ah$ ($P$ lies outside the parabola). Then

$$t_1 + t_2 = \frac{k}{a}, \qquad t_1 t_2 = \frac{h}{a}.$$




The tangent with parameter $t$ meets the directrix $x = -a$ where $ty = -a + at^2$. Provided $t \neq 0$,

$$y = a\left(t - \frac{1}{t}\right).$$

(If $t = 0$ the tangent is $x = 0$, parallel to the directrix, and there is no point of intersection. That happens only when $h = t_1 t_2\, a = 0$, so for $L$ and $M$ to exist we need $h \neq 0$, and then $t_1, t_2 \neq 0$.) So $L$ and $M$ are $(-a,\ a(t_1 - \frac{1}{t_1}))$ and $(-a,\ a(t_2 - \frac{1}{t_2}))$, and since both lie on the same vertical line,

$$\begin{aligned} LM &= \left|a(t_1 - t_2) - a\left(\frac{1}{t_1} - \frac{1}{t_2}\right)\right| = \left|a(t_1 - t_2) + a\,\frac{t_1 - t_2}{t_1 t_2}\right| \\ &= |a|\,|t_1 - t_2|\,\left|\frac{t_1 t_2 + 1}{t_1 t_2}\right|. \end{aligned}$$

**Ask yourself** How do we get $t_1 - t_2$ without solving the quadratic?

Its square is symmetric in the roots:

$$(t_1 - t_2)^2 = (t_1 + t_2)^2 - 4t_1 t_2 = \frac{k^2}{a^2} - \frac{4h}{a} = \frac{k^2 - 4ah}{a^2}.$$

Also

$$\frac{t_1 t_2 + 1}{t_1 t_2} = \frac{h/a + 1}{h/a} = \frac{h + a}{h}.$$

Squaring the expression for $LM$ and substituting,

$$LM^2 = a^2 \cdot \frac{k^2 - 4ah}{a^2} \cdot \frac{(h+a)^2}{h^2} = \frac{(h+a)^2(k^2-4ah)}{h^2}.$$

If $LM = \ell$, multiplying through by $h^2$ gives

$$(h+a)^2(k^2-4ah) = \ell^2 h^2,$$

so $P = (h,k)$ lies on $(x+a)^2(y^2-4ax) = \ell^2 x^2$, as required.

## Checks

- Take $a = 1$ and $P = (1,3)$, as in the diagram. The parameters satisfy $t^2 - 3t + 1 = 0$, so $t = \frac{1}{2}(3 \pm \sqrt{5})$, and $t - \frac{1}{t} = \pm\sqrt{5}$ (because $t_1 t_2 = 1$ makes $1/t_1 = t_2$). So $LM = 2\sqrt{5}$ and $LM^2 = 20$, while the formula gives $(1+1)^2(9-4)/1^2 = 20$.
- A point on the directrix, $h = -a$: the formula says $LM = 0$. Indeed the tangents from such a point meet the directrix at the point itself, so $L = M$.
- We found the tangents independently (lines $y = k + m(x-h)$ with $hm^2 - km + a = 0$) for over 14,000 random $a$, $h$, $k$, and the computed $LM^2$ matched $(h+a)^2(k^2-4ah)/h^2$ every time. For example $a = 1$, $P = (2,5)$ gives $LM^2 = 38.25$ both ways.

## Common slips

- Using the tangent's parameter equation the wrong way round: $P$ fixed and $t$ unknown gives the quadratic $at^2 - kt + h = 0$, not one in $h$ and $k$.
- Writing $\frac{1}{t_1} - \frac{1}{t_2} = \frac{t_1 - t_2}{t_1 t_2}$; it is $\frac{t_2 - t_1}{t_1 t_2}$, and the sign is what makes the two terms add rather than cancel.

## Beyond STEP

The tangents from a point to a conic, and the line joining their points of contact (the *polar* of the point), are the start of projective geometry, where poles, polars and the directrix fit into one picture: the directrix is the polar of the focus. Working with roots through their symmetric functions, as here, is also the first step towards the theory of equations and Galois theory. Siklos's *Advanced Problems in Mathematics* has more coordinate geometry problems in this style.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.